

9 Energy stored in Capacitors and Electric Fields; Dielectrics

9.1 Lecture Notes

- Charge goes up (conceptual example).
- Lets you apply higher voltages (so more charge).
- Lets you place the plates closer together (make d smaller).
- Increases the value of C because $\kappa > 1$.

9.2 Recitation

- Before the exam, we had a capacitance equation with a κ .
 - Suppose we disconnect from battery, so Q is constant.
 - Now put an insulator inside capacitor.
 - * Tada, there's your κ
 - * Insulator means electrons are stuck where they're at.
 - * $\kappa > 1$, usually.
 - This causes an electric field inside, causes a polarization.
- We can express the energy in three different forms, depending on what we know about the system.
- Energy conservation is inapplicable to these problems.